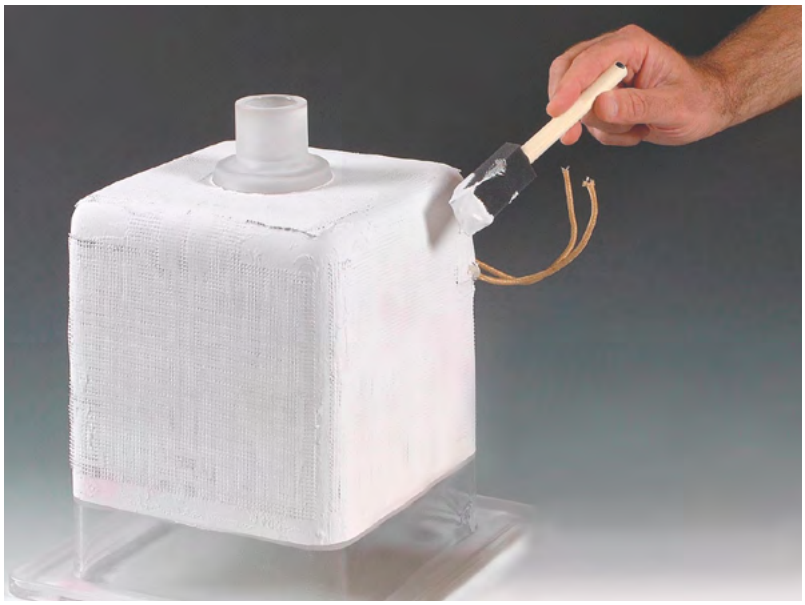
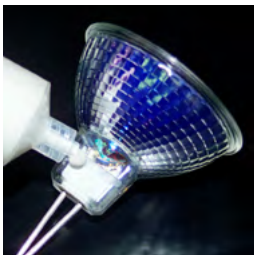
AREMCO

HIGH TEMPERATURE CERAMIC ADHESIVES

Technical Bulletin A2-S1



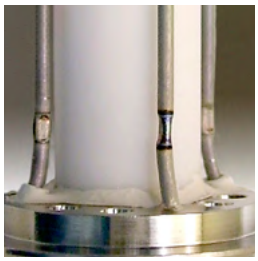
Ceramabond™ 569 bonds flex heater to quartz vessel.



Ceramabond™ 835-M bonds halogen lamp.

Ceramabond™ 503 coats heater used to 1700 °C.

Ceramabond™ 503 coats spiral cantilevered sensor.



Ultra-Temp™ 516 seals heater assembly.

Aremco's high temperature ceramic adhesives are formulated using a broad range of ceramics fillers and inorganic binders, and are ideal for bonding, potting and sealing ceramics, composites, graphite, refractory metals, quartz, and semiconductors for applications to 3200 °F (1760 °C).

Part No.	Filler	Bonding*	Principal Use
503	Al_2O_3	C–C	Dense Ceramics; Alumina-to-Alumina
520		C–C	High Purity Alumina Fiberboard & Refractories
552		C–C, C–M	Solid Oxide Fuel Cells; Low CTE Metals
569		C–C, C–M, Quartz	Probes, Sensors, Resistors, Igniters, Heaters
670		C–C, C–M	Ceramic Textiles, Thread-Locking
671		C–C, C–M, M–M	Ceramic Textiles, Thread-Locking
835-M		C–C, C–M, Quartz	Halogen Lamps
865	AlN	C–C, C–M	Probes & Sensors; Thermal Conductivity
600-N	$Al_2O_3 – SiO_2$	C–C, C–M	Refractory Repair
571	MgO	C–M, M–M	Heaters, Induction Coils, Sensors
618-N	SiO_2	C–C, Quartz	Porous Ceramics, Quartz Tubes & Vessels
516	$ZrO_2 – ZrSiO_4$	C–C, C–M	Thermocouples, Semiconductor Wafers
835		C–C, C–M	Halogen Lamps
885	ZrO_2	C–C	Zirconia, Solid Oxide Fuel Cells
885-K		C–M	
890	SiC	C–C	Crucibles, Heaters, Sagger Plates
890-K		C–M	Probes, Sensors, Infrared Heaters

*C–C = Ceramic-to-Ceramic C–M = Ceramic-to-Metal M–M = Metal-to-Metal

TYPICAL APPLICATIONS

Electrical

- Halogen Lamps
- Heaters
- Igniters
- Fiberoptics
- Resistors
- Solid Oxide Fuel Cells

Instruments & Sensors

- Gas Chromatographs
- High Vacuum Components
- Liquid Metal Inclusion Counters
- Mass Spectrometers
- Oxygen Analyzers
- Strain Gauges
- Semiconductors
- Temperature Probes

Mechanical

- Ceramic Honeycombs
- Ceramic Textiles
- Graphite Blocks
- Refractory Insulation
- Sagger Plates
- Thread-Locking

CERAMABOND™ — HIGH TEMPERATURE CERAMIC ADHESIVES PROPERTIES

Part Number		503	520	552	569	670	671	835-M	600-N	865
Major Constituent					Al_2O_3				$Al_2O_3 – SiO_2$	AlN
Color		White	White	White	White	White	White	White	Tan	Gray
Temperature Limit, °F (°C)		3000 (1650)	3000 (1650)	3000 (1650)	3000 (1650)	3000 (1650)	3200 (1760)	3000 (1650)	3000 (1650)	3000 (1650)
No. Components		1	1	1	1	1	1	1	1	1
Viscosity, cP		50,000–90,000	1,000–3,000	53,000–73,000	Paste	2,500–5,000	40,000–80,000	30,000–40,000	5,000–15,000	Paste
Specific Gravity, g/cc		2.35–2.55	2.60–2.65	1.90–2.20	2.15–2.30	1.80–1.95	2.05–2.15	2.35–2.45	2.00–2.05	1.95–2.15
CTE, in/in/°F × 10^{-6} (°C)		4.0 (7.2)	4.0 (7.2)	4.3 (7.7)	4.2 (7.6)	4.3 (7.7)	4.3 (7.7)	4.0 (7.2)	3.0 (5.4)	1.5 (2.7)
Handling	**Mix Ratio,** powder:liquid	NA	NA	NA	NA	NA	NA	NA	NA	NA
Handling	**Thinner**	503-T	520-T	552-T	569-T	670-T	671-T	835-M-T	600-T	865-T
Handling	**Solvent**	Water	Water	Water	Water	Water	Water	Water	Water	Water
Handling	**Application Temperature,** °F	50–90	50–90	50–90	50–90	50–90	50–90	50–90	50–90	50–90
Handling	**Storage Temperature,** °F	40–90	40–90	40–90	40–90	40–90	40–90	40–90	40–90	40–90
Handling	**Shelf Life,** months	6	6	6	6	6	6	6	6	6
Curing	**Air Set,** hrs	≤ 1	1–2	1–4	1–4	1–4	1–4	1–4	1–4	1–4
Curing	**Heat Cure,** °F, hrs	200, 2 + 500, 2 + 700, 2	200, 2 + 800, 1	200, 2 + 500, 2	200, 2	200, 2	200, 2	200, 2	200, 2 + 350, 1	200, 2 + 350, 2 + 500, 2
Dielectric Strength, volts/mil @ RT		171	171	173	138	142	182	163	203	187
Torque Strength, ft-lbs[1]		60	10	52	38	60	57	63	14	27
Moisture Resistance[2]		Good	Good	Excellent	Excellent	Excellent	Excellent	Good	Excellent	Excellent
Alkali Resistance[2]		Fair	Good	Good	Good	Good	Excellent	Excellent	Good	Good
Acid Resistance[2]		Excellent	Good	Good	Excellent	Good	Good	Good	Good	Good

Footnotes

[1] Tested using a torque wrench after bonding a pre-oxidized ½"–13 nut and bolt and final curing at 1000 °F.

[2] Properties were evaluated after curing at 700 °F for 2 hours.

General Notes

1. Ceramabond adhesives do not contain volatile organic compounds (VOCs).
2. Special pigments available upon request.
3. Many adhesives including 503, 516, 552, 569, 571, 618-N, 671, 835-M, and 890 can be formulated using 1-5 micron ceramic powders. Add "-VFG" to the part number (eg. 503-VFG).

Abbreviations

NA Not Applicable
NM Not Measured

CERAMABOND™ — HIGH TEMPERATURE CERAMIC ADHESIVES PROPERTIES

Part Number		571	618-N	890	890-K	516	835	885	885-K
Major Constituent		MgO	SiO_2	SiC		ZrO_2 – $ZrSiO_4$		ZrO_2	
Color		Off-White	Off-White	Blue-Gray	Blue-Gray	Tan	Tan	Tan	Tan
Temperature Limit, °F (°C)		3200 (1760)	3000 (1650)	3000 (1650)	3000 (1650)	3200 (1760)	3000 (1650)	3200 (1760)	3200 (1760)
No. Components		2	1	1	1	1	1	1	1
Viscosity, cP		20,000–90,000[3]	40,000–60,000	35,000–55,000	10,000–40,000	40,000–70,000	20,000–40,000	10,000–20,000	10,000–30,000
Specific Gravity, g/cc		1.90–2.20	1.80–1.90	1.70–1.75	2.35–2.45	2.15–2.30	2.25–2.35	2.65–2.70	2.65–2.70
CTE, in/in/°F × 10^{-6} (°C)		7.0 (12.6)	.33 (.59)	2.4 (4.4)	3.0 (5.4)	4.1 (7.4)	4.0 (7.2)	4.0 (7.2)	4.2 (7.6)
Handling	**Mix Ratio,** powder:liquid	1.0:1.0, 1.5:1.0	NA	NA	NA	NA	NA	NA	NA
Handling	**Thinner**	571-T	618-N-T	890-T	890-K-T	516-T	835-T	885-T	885-K-T
Handling	**Solvent**	Water	Water	Water	Water	Water	Water	Water	Water
Handling	**Application Temperature,** °F	50–90	50–90	50–90	50–90	50–90	50–90	50–90	50–90
Handling	**Storage Temperature,** °F	40–90	40–90	40–90	40–90	40–90	40–90	40–90	40–90
Handling	**Shelf Life,** months	6	6	6	6	6	6	6	6
Curing	**Air Set,** hrs	1–4	1–4	≤ 1	1–4	1–4	≤ 1	≤ 1	1–4
Curing	**Heat Cure,** °F, hrs	200, 2	200, 2 + 500, 2 + 700, 2	200, 2 + 500, 2 + 700, 2	200, 2 + 500, 2	200, 2 + 500, 2 + 700, 2	200,2	200, 2 + 500, 2 + 700, 2	200, 2 + 500, 2
Dielectric Strength, volts/mil @ RT		91	156	73	95	188	111	105	88
Torque Strength, ft-lbs[1]		22	77	40	50	50	50	40	20
Moisture Resistance[2]		Excellent	Excellent	Good	Good	Good	Good	Good	Good
Alkali Resistance[2]		Good	Good	Good	Good	Excellent	Good	Good	Good
Acid Resistance[2]		Fair	Good	Good	Good	Good	Good	Good	Good

Footnotes

[1] Tested using a torque wrench after bonding a pre-oxidized ½"–13 nut and bolt and final curing at 1000 °F.

[2] Properties were evaluated after curing at 700 °F for 2 hours.

[3] Ceramabond™ 571 ranges for viscosity and specific gravity reflect a powder-to-liquid mix ratio that ranges from 1-to-1 to 1.5-to-1.

General Notes

1. Ceramabond adhesives do not contain volatile organic compounds (VOCs).
2. Special pigments available upon request.
3. Many adhesives including 503, 516, 552, 569, 571, 618-N, 671, 835-M, and 890 can be formulated using 1-5 micron ceramic powders. Add "-VFG" to the part number (eg. 503-VFG).

Abbreviations

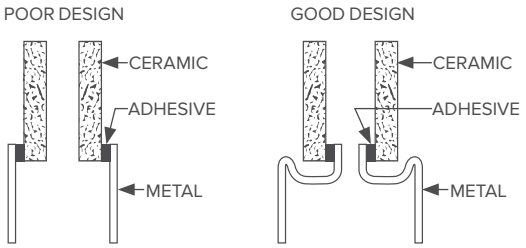
NA Not Applicable
NM Not Measured

DESIGN GUIDELINES

General design criteria for bonding with ceramic adhesives are similar to those for epoxy adhesives. Main considerations include the **coefficient of thermal expansion, joint design, glue line thickness,** and **operating environment.**

Coefficient of Thermal Expansion

CERAMIC-TO-METAL RECOMMENDED DESIGN

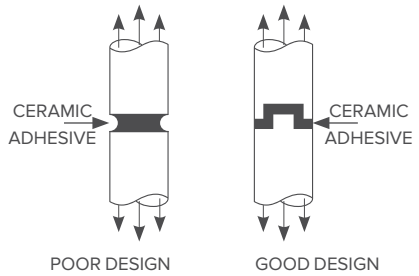


Due to the high thermal loading implicit in most ceramic adhesive applications, the joint design should account for the difference in the coefficient of thermal expansion between the adhesive and the components that are being joined. In the illustration above, note that the "poor" design loads the adhesive in tension since the metal expands faster than the ceramic. The "good" design allows for this thermal mismatch and loads the adhesion in compression, offering higher reliability.

Joint Design

Most adhesives offer relatively poor tensile-shear strength, so it is important to design a joint that will distribute the mechanical stress by maximizing the length of the glue line as shown in this illustration.

CERAMIC-TO-CERAMIC RECOMMENDED JOINT DESIGN



Glue Line Thickness

The clearance between mating parts at operating temperature should be 2–8 mils (50–200 microns). Less than 2 mils will prevent uniform adhesion; greater than 8 mils will often result in cohesive shear failure within the adhesive. A maximum depth of 0.25″ is recommended when using a ceramic adhesive for a small potting application.

Operating Environment

These adhesives offer excellent chemical, electrical and ultra high thermal resistance, and do not outgas under high vacuum. The main limitations are (a) relatively low mechanical strength and (b) slight porosity after curing. Contact Aremco for suggestions about how to reduce porosity and produce gas-tight seals.

APPLICATION PROCEDURES

Surface Preparation

Smooth surfaces are difficult to bond and should be etched, abrasive blasted or oxidized, then cleaned thoroughly prior to application. Aremco's Corr-Prep™ CPR2000 is recommended for etching metals. Porous substrates should be pre-coated with a binder (thinner) to prevent separation and absorption of the adhesive binder. Add a "-T" to the part number (eg. 503-T) to designate the product thinner.

Mixing

One-part adhesives tend to settle and should be mixed thoroughly prior to use. Refer to Tech Bulletin A12 for information about Aremco's **Model 7000 Pneumatic Mixer.** Mix ratios for two-part adhesives are shown in the Property Chart. Viscosity may be adjusted by thinning up to 20% by weight.

Application

Apply a thin coat of adhesive to each surface using a brush, spatula or dispenser. Using a clamp or similar tool, maintain a uniform glue line of 2–8 mils (50–200 microns) by applying even pressure across the assembly. Wipe away excess material prior to drying. Refer to Tech Bulletin A12 for optional dispensing tools.

Model 7000 Mixer

Curing

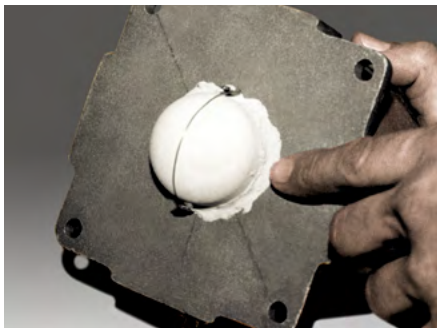
Refer to the Property Chart for specific curing instructions for each product. A ramp rate of ~5 °F per minutes is recommended. If any blistering occurs, repeat curing process by slowing down the ramp rate. Blistering may be caused by an excessively quick release of water vapor from the adhesive.

Note that Ceramabond adhesives cure by dehydration. Although the primary outgassing by-product is water vapor, small amounts of inorganic binder and ceramic filler may also attach to the water vapor during the curing cycle. Although most of the dehydration occurs at 200–350 °F, chemically absorbed water may remain until the adhesive is exposed to temperatures in excess of 700–800 °F.

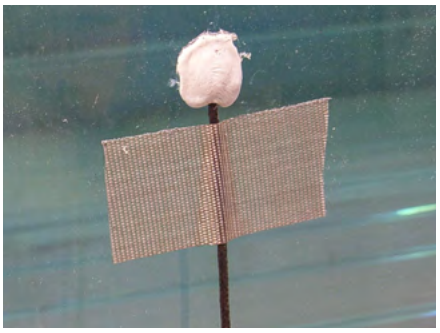
Cleanup

Uncured material can be cleaned with warm water and soap. Cured material can often be removed using warm water and ultrasonics. Cured material that has been cycled to ultra-high temperatures is more difficult to remove; it can be softened sometimes with warm water and ultrasonics, but a strong acid such as HCl may be required for removal. Mechanical abrasion may also be necessary.

Ceramabond™ 571 coats copper induction heater.



Ceramabond™ 571 coats oxygen sensor.

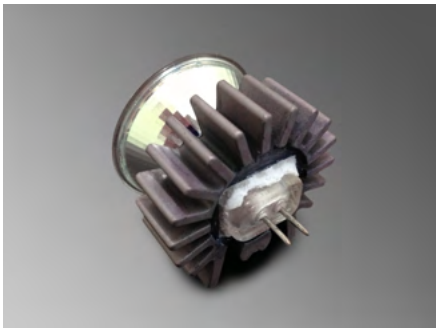


Ceramabond™ 571 bonds thermocouple to glass.

Ceramabond™ 618-N bonds porous ceramic filter elements.

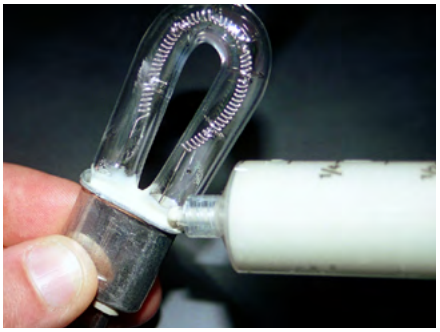
Ceramabond™ 671 used as a high temp threadlocker.

Ceramabond™ 503 repairs furnace saggar plate.



Ceramabond™ 835-M bonds heat sink to halogen lamp.

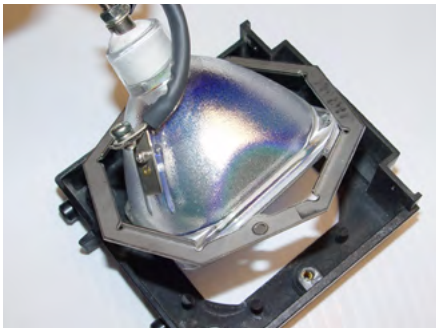
Ultra-Temp™ 516 bonds thermocouple to quartz tube.



Ceramabond™ 835 bonds halogen lamp.

Ceramabond™ 552 seals thermocouple in metal housing.

Ceramabond™ 835-M bonds cover to halogen lamp.



Ceramabond™ 835-M bonds halogen lamp.

CERAMIC ADHESIVE SELECTOR CHART

	Material	CTE °F (°C)	503	520	552	569	670	671	835-M	600-N	865	571	618-N	890	890-K	516	835	885	885-K
			Al_2O_3							Al_2O_3 – SiO_2	AlN	MgO	SiO_2	SiC		ZrO_2 – $ZrSiO_4$		ZrO_2	
CERAMICS	Alumina	4.4 (7.9)	•	•	•	•	•	•	•			x							
CERAMICS	Alumina-Silica	1.8 (3.2)								x									
CERAMICS	Aluminum Nitride	1.5 (2.7)									•		x						
CERAMICS	Beryllia	4.1 (7.4)	•		x	x	x	x	x							x	x	x	
CERAMICS	Boron Carbide	2.6 (4.7)	x											x					
CERAMICS	Boron Nitride	4.2 (7.6)	x																
CERAMICS	Borosilicate Glass	1.8 (3.2)	x										•						
CERAMICS	Calcium Silicate	3.0 (5.4)					•												
CERAMICS	Ceramic Textile	–					•	x											
CERAMICS	Cordierite	1.1 (2.0)											•						
CERAMICS	Graphite	4.3 (7.7)	x											x					
CERAMICS	Macor	5.2 (9.4)			x	•	x	x	x			x							
CERAMICS	Mica	4.7 (8.5)																	
CERAMICS	Mullite	3.0 (5.4)	x		x	x	x									x	x		
CERAMICS	Quartz	0.30 (0.54)	x			x			x				•				x		
CERAMICS	Refractory, Dense	–	•	•										•	•			x	
CERAMICS	Refractory, Porous	–		•						•									
CERAMICS	Sapphire	4.2 (7.6)	•			x	x		x										
CERAMICS	Silica	0.31 (0.56)											•						
CERAMICS	Silicon Carbide	2.9 (5.2)	x											•	•				
CERAMICS	Silicon Nitride	1.8 (3.2)	x								x		x	x					
CERAMICS	Steatite	4.0 (7.2)			x	•		x	x								•		
CERAMICS	Zirconia	5.7 (10.3)														x	x	•	•
CERAMICS	Zirconia Silicate	4.0 (7.2)														•	•	x	
METALS	Aluminum	15.0 (27.0)										•							
METALS	Brass	10.2 (18.4)										•							
METALS	Cast Iron	5.9 (10.6)			x	x	x	x	x			•							
METALS	Copper	9.3 (16.7)										•							
METALS	Inconel	6.4 (11.5)			x	x	x	x	x			•							
METALS	Molybdenum	2.9 (5.2)			x	•	x	x	x							x	x		
METALS	Nickel	7.2 (13.0)										•							
METALS	Nickel-Iron	2.6 (4.7)			x	•	x	x	x							x	x		
METALS	Platinum	4.9 (8.8)	•		x	x	x												
METALS	Silicon	1.6 (2.9)									x					x	x		
METALS	Silver	10.6 (19.1)										x							
METALS	Stainless (300 Series)	9.6 (17.3)										x			x				x
METALS	Stainless (400 Series)	6.2 (11.2)			x	x	x	x	x			•			x	x	x		x
METALS	Steel (1010)	6.5 (11.7)			x	x	x	x	x			•			x	x	x		x
METALS	Tantalum	3.9 (7.0)			x	x	x	x	x			x			x	x	x		
METALS	Titanium	5.8 (10.4)			x	x	x	x	x			•			x	x	x		x
METALS	Tungsten	2.5 (4.5)			x	•	x	x	x							x	x		

• = Preferred, x = Applicable

Refer to Price List for availability of sample kits and complete order information.

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The user assumes all risk of use or handling whether or not in accordance with directions or suggestions, or used singly or in combination with other products.